

Lecture Notes: Linear Systems, Determinants, and Ill-Conditioning

1. Solving $A\mathbf{x} = \mathbf{b}$ with the inverse

A linear system can be written $A\mathbf{x} = \mathbf{b}$. If A has an inverse, multiply both sides by it:

$$\mathbf{x} = A^{-1}\mathbf{b}$$

This is the matrix version of dividing both sides of $ax = b$ by a . The warm-up system $x + y = 10$, $x - y = 2$ is $A = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$, and $A^{-1}\mathbf{b}$ gives $(6, 4)$.

2. The determinant is what we divide by

Every formula for the inverse divides by $\det(A)$. For a 2x2,

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

and Cramer's rule says $x_i = \det(A_i)/\det(A)$, where A_i is A with column i replaced by $\mathbf{b}$. If $\det(A) = 0$ we are dividing by zero, so there is no inverse and no unique solution. Depending on $\mathbf{b}$, the system has either no solution (our $x + y = 10$, $x + y = 12$) or infinitely many (change the 12 to a 10).

3. When is the determinant zero?

The easy cases: two rows are identical, or one row is a constant multiple of another ($r_1 = cr_2$). In the warm-up, system 2 has two identical rows on the left, and in system 3 row 3 is exactly $2 \cdot r_1$. Geometrically, two of the equations describe the "same direction" and so the rows carry less information than it looks like.

4. Linear combinations of rows

Rows do not have to be copies of each other. Take

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 5 & 7 & 9 \end{pmatrix}$$

Here $r_1 + r_2 = r_3$. Expanding along the first row,

$$\det(A) = 1(45 - 42) - 2(36 - 30) + 3(28 - 25) = 3 - 12 + 9 = 0.$$

The general principle: if some combination of the rows adds up to the zero row, that is $c_1 r_1 + c_2 r_2 + c_3 r_3 = \mathbf{0}$ with the c_i not all zero, then $\det(A) = 0$. Here the combination is $r_1 + r_2 - r_3 = \mathbf{0}$. Section 3 is the special case where the combination involves only two rows.

5. The homogeneous test

The statement above can be read as an equation. A combination of rows is $\mathbf{x}^T A$, so “some nontrivial combination of rows is zero” says

$$\mathbf{x}^T A = \mathbf{0}^T \text{ has a nonzero solution } \mathbf{x}.$$

(For the example, $\mathbf{x} = (1, 1, -1)$.) The same is true for columns: if $A\mathbf{x} = \mathbf{0}$ has a nonzero solution (here $\mathbf{x} = (1, -2, 1)$), then $\det(A) = 0$. This is the cleanest way to think about it: the determinant is zero exactly when there is some nonzero input that the matrix sends to zero, and information is lost. A matrix that loses information cannot be undone.

6. Close to zero is also bad

A determinant that is zero is a disaster; a determinant that is merely very small is a quieter disaster. The inverse still exists, but we are dividing by a tiny number, so small errors in $\mathbf{b}$ (measurement noise, rounding) get blown up in $\mathbf{x}$. Such a system is called ill-conditioned. This is the subject of the ill-conditioning handout.

7. Why this matters in machine learning

Suppose we fit a linear model by solving the normal equations $(X^T X)\mathbf{w} = X^T \mathbf{y}$. Solving this means inverting $X^T X$. If two features (columns of X) are collinear, or nearly so (a very high Pearson r), then $X^T X$ is singular or nearly singular, and we are back in sections 4 through 6. The weights become huge, unstable, and swing wildly with small changes in the data.

A practical remedy is to drop one of the redundant features, since it carries almost no information the other does not already carry. Since $\det(A) = \det(A^T)$, everything above applies equally to rows and columns, so “dependent rows” and “dependent columns” are the same problem.